Research Proposal: Multimodal Learning-Disorder Screening

Title

Multimodal Learning-Disorder Screening Using Handwriting, Speech, Text, and Reading-Behavior Signals

Problem Statement

Single-signal screening can miss important educational evidence. This project already combines handwriting, speech, text, and behavior inputs, which makes it a strong base for a multimodal learning-disorder screening study.

Research Scope

- compare baseline models against multimodal fusion models
- test whether attention-based fusion improves interpretability
- evaluate whether a ViT handwriting branch improves image sensitivity
- compare binary risk, severity classification, and regression-style scoring
- select the best model with cross-validation, then sanity-check it on a hard holdout split
- compare old and new threshold settings side by side when recall needs tuning

Hypothesis

Multimodal fusion will outperform single-modality baselines, and attention fusion will improve the usefulness of explanations.

Data Requirements

- anonymized manifest CSV
- handwriting images
- reading audio files
- text samples
- spelling and pronunciation counts
- behavior features

Candidate Methods

- InitialCNNModel
- InitialLSTMModel

- InitialCNNLSTMModel
- MultimodalDyslexiaModel
- TransformerMultimodalModel
- AttentionMultimodalModel

Evaluation Ideas

- accuracy
- F1 score
- MAE for regression mode
- calibration / confidence analysis
- modality-attention consistency
- validation matrix reporting
- holdout test sanity-check

Expected Output

- best-performing screening architecture
- ranked evidence sources
- teacher-friendly explanation summary

Main Risks

- small dataset size
- noisy labels
- class imbalance
- modality missingness